INCH-POUND

MIL-DTL-32262A (MR)
16 March 2015
SUPERSEDING
MIL-DTL-32262 (MR)
31 July 2007

DETAIL SPECIFICATION

# ARMOR PLATE, ALUMINUM ALLOY 6055 WELDABLE & ALLOY 6061, UNWELDABLE APPLIQUE

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

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## 1. SCOPE

1.1 Scope. This specification covers two wrought aluminum alloy armor plate for both welded and non-welded applications in nominal thicknesses from 0.250 to 3.000 inch, inclusive (see 6.2). The weldability of wrought aluminum alloy AA6055 armor has been determined for Class II armor for this thicknesses, only; 1.500". The weldability of AA6061 wrought aluminum alloy armor has not been determined. Therefore, this alloy should be used as appliqué armor.

1.2 Weldability. Class II material covered by this specification has been demonstrated to be weldable to itself and other weldable alloys (see 6.3 and 6.4).

1.3 Classification. The wrought aluminum armor should be of the following classes as specified (see 6.2).

1.3.1 Class I. Class I is wrought aluminum armor that conforms to the Aluminum Association designation for the 6061 aluminum alloy. The applicable gauge range for Class I is 0.250-3.000 inches.

1.3.2 Class II. Class II is wrought aluminum armor that conforms to the Aluminum Association designation for the 6055 aluminum alloy. The applicable gauge range for Class II is 0.500-3.000 inches.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army Research Laboratory, Weapons and Materials Research Directorate, Materials and Manufacturing Technology Branch, Specifications and Standards Office, Attn: RDRL-WMM-D, Aberdeen Proving Ground, MD 21005-5069 or emailed to richard.j.squillacioti.civ@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 9535

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## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

| | | |
|---|---|---|
| MIL-STD-662 | - | $V_{50}$ Ballistic Test for Armor |

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal should apply.

THE ALUMINUM ASSOCIATION, INC.

| | | |
|---|---|---|
| ANSI H35.2 | - | American National Standard Dimensional Tolerances for Aluminum Mill Products |

(Copies of these documents are available online at http://www.aluminum.org).

ASTM INTERNATIONAL

| | | |
|---|---|---|
| ASTM B557 | - | Standard Test Methods for Tension Testing Wrought and Cast Aluminum and Magnesium-Alloy Products |
| ASTM E34 | - | Standard Test Methods for Chemical Analysis of Aluminum and Aluminum-Base Alloys |
| ASTM E716 | - | Standard Practices for Sampling Aluminum and Aluminum Alloys for Spectrochemical Analysis |
| ASTM E1251 | - | Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Atomic Emission Spectrometry |

(Copies of these documents are available online at http://www.astm.org.)



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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), first article testing shall be required and all test samples required by this specification shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3. First article testing shall be completed before production material is submitted for acceptance testing. The approval of the first article samples authorizes commencement of production but does not relieve the supplier of the responsibility to comply with all the applicable provisions of this specification. The first article samples and acceptance test plates shall be manufactured by the process proposed for use on production items.

3.2 Chemical composition. The chemical composition of the plates shall be within the limits shown in Table I. The limits specified in Table I were taken from Teal Sheets from the Aluminum Association dated 2009-03-01 registered composition limits for Aluminum Alloys. A certification of conformance of the chemical composition of the alloy shall be furnished with the ballistic test plates.

TABLE I. Chemical composition, percent. [1/]

| ELEMENTS | SYMBOL | Class I 6061 ALLOY [2/] | Class II 6055 ALLOY [2/] |
|---|---|---|---|
| Silicon | Si | 0.40 – 0.80 | 0.60 - 1.2 |
| Iron | Fe | 0.70 | 0.30 |
| Copper | Cu | 0.15 – 0.40 | 0.50 - 1.0 |
| Manganese | Mn | 0.15 | 0.10 |
| Magnesium | Mg | 0.8 – 1.2 | 0.70 - 1.1 |
| Chromium | Cr | 0.04 – 0.35 | 0.20 - 0.30 |
| Zinc | Zn | 0.25 | 0.55 - 0.90 |
| Titanium | Ti | 0.15 | 0.10 |
| Other, max. Each | --- | 0.05 | 0.05 |
| Other, max. Total [3/] | --- | 0.15 | 0.15 |
| Aluminum | Al | Remainder | Remainder |

[1/] Except for “Aluminum” and “others”, analysis normally is made for elements for which specific limits are shown.
[2/] Where single units are shown, these indicate the maximum amounts permitted.
[3/] The sum of those “others” metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum.

3.3 Mechanical properties. Unless otherwise specified in the contract or order (see 6.2), the mechanical properties of the test specimen taken in the longitudinal (L) direction shall meet the minimum mechanical properties listed in Table II. If mechanical property requirements differ from those contained in Table II, or if any other properties are required, the ballistic requirements shall be negotiated between the procuring activity and the supplier.



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TABLE II. Minimum mechanical properties. 1/ 2/

| Thickness, inches | Tensile Strength, Ksi | | Yield Strength, 0.2% Offset, Ksi | | Elongation percent | |
|---|---|---|---|---|---|---|
| | Class I | Class II | Class I | Class II | Class I | Class II |
| 0.250 to 0.499, incl. | 38.0 | N/A | 35.0 | N/A | 10 | N/A |
| 0.500 to 2.000, incl. | 38.0 | 52 | 37.0 | 50 | 10 | 6 |
| 2.001 to 3.000, incl. | 38.0 | 52 | 35.0 | 50 | 10 | 5 |

1/ The test specimen gage length shall be 1.400 inch for plates having a nominal thickness of 0.500 inch; all other thicknesses shall have a gage length of 2.000 inches.

2/ Values are taken in the Longitudinal Direction (see 3.3).

3.4 Weldability. When the vehicle requires weldments, the fabricator (OEMs and kit manufacturers, etc.) are required to demonstrate weldability (which includes ballistic shock testing) for armor to armor and armor to non-armor applications in accordance with the Ground Combat Vehicle Welding Code (GCVWC) - Aluminum #19207-12472301. Since the alloys referenced by this specification are not specified in the GCVWC 12472301 or in the revised Military Standard for Weldability of Aluminum Alloys (MIL-STD-1946) it is strongly suggested that the contract or drawing for the vehicle specify weldability requirements as outlined in paragraph 6.3. Aluminum fabricators shall qualify a weld procedure for any new armor material being used for armor applications and this requires ballistic shock testing. This requirement (ballistic shock testing and the associated striking velocities) are listed in Appendix A and shall also be specified in the contract or drawing for the vehicle, if the alloy specified by this specification is to be welded at the following thicknesses; Class II material in the thicknesses of 1.500".

3.5 Ballistic limit. The protection ballistic limit, BL(P), shall be as specified in Appendix B. No ballistic limit for Class I material shall be specified for plate thickness of 0.250 to 0.499 inch, inclusive. When a complete penetration cannot be obtained for any class of armor material, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a $V_{50}$ (which obviously cannot be explicitly determined) above the minimum requirement.

3.6 Thermal processing. Any thermal processing exceeding 200°F (93°C) excluding fabrication by welding shall necessitate recertification for conformance to 3.3 and 3.4.

3.7 Dimensions. Dimensions for plates delivered for fabrication shall have an overall dimension tolerance of +0.500/-0.000 for width and length unless otherwise specified in the contract or order (see 6.2).



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3.7.1 Tolerances. Unless otherwise specified in the contract or order (see 6.2), the plates shall not vary from the specified ordered dimensions by an amount greater than that specified by ANSI H35.2, except for thickness.

3.7.2 Thickness. Thickness tolerance for production armor and ballistic test plates shall be as specified in Table III.

TABLE III. Thickness tolerances.

| Ordered Thickness (Inches) | | SPECIFIED WIDTH (Inches) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | OVER | 0.00 | 39.37 | 59.06 | 78.74 | 98.43 | 118.11 | 137.80 | 157.48 |
| | | THRU | 39.37 | 59.06 | 78.74 | 98.43 | 118.11 | 137.80 | 157.48 | 177.17 |
| OVER | THRU | TOLERANCES - INCHES (PLUS and MINUS) | | | | | | | | |
| 0.250 | 0.315 | | 0.012 | 0.014 | 0.015 | 0.018 | 0.022 | 0.027 | 0.035 | 0.043 |
| 0.315 | 0.394 | | 0.015 | 0.017 | 0.020 | 0.023 | 0.027 | 0.033 | 0.041 | 0.051 |
| 0.394 | 0.630 | | 0.023 | 0.023 | 0.027 | 0.032 | 0.035 | 0.043 | 0.053 | 0.065 |
| 0.630 | 0.984 | | 0.031 | 0.031 | 0.037 | 0.043 | 0.047 | 0.058 | 0.070 | 0.085 |
| 0.984 | 1.575 | | 0.039 | 0.039 | 0.047 | 0.055 | 0.065 | 0.075 | 0.090 | 0.105 |
| 1.575 | 2.362 | | 0.055 | 0.055 | 0.060 | 0.070 | 0.085 | 0.100 | 0.115 | --- |
| 2.362 | 3.000 | | 0.075 | 0.075 | 0.085 | 0.100 | 0.105 | 0.125 | --- | --- |

3.8 Marking for identification. Unless otherwise specified in the contract or purchase order (see 6.2) each plate shall be marked on one plate edge with the manufacturer's name or CAGE code, the basic number of this specification, the plate thickness in inches, the alloy designation and the lot number or code relating to the lot number (see 4.2). The height of the characters shall be 3/8 of an inch or greater. Impression stamping shall not be used unless permitted by the procuring activity (see 6.2). Each plate shall be marked in lengthwise rows of characters recurring at intervals not greater than 3 feet, the rows being spaced not more than 6 inches apart and alternately staggered. The characters shall be not less than 3/8 inch in height and shall be applied using a suitable marking fluid whose residue shall not contain more than traces of halogen-bearing compounds and shall be capable of being removed in hot alkaline cleaning solution without rubbing. The markings shall have no deleterious effect on the plate material or its performance and shall be sufficiently stable to withstand normal handling.

3.8.1 Ballistic test plates. In addition to the markings in 3.8, each ballistic test plate shall be marked with the letters PRE for First Article test plates and ACC for Acceptance test plates. This marking shall be impression stamped on the edge in letters 3/8 inch high or greater. Ballistic retest plates shall be marked "R1" and "R2" respectively (see B.5.2).

3.9 Ballistic test plate information. For each lot of aluminum alloy armor a properly completed Aluminum Armor Test Data Form (See Figure 1) shall be submitted with each ballistic test plate that represents that particular processing lot.

3.10 Workmanship. Plate produced under this specification shall be uniform in quality and clean, smooth and sufficiently free from buckles, blisters, hard spots, damaged ends, laminations and other defects which may affect its use.



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| REQUEST FOR BALLISTIC TEST OF ALUMINUM ALLOY 6061/ 6055 ARMOR | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **FIRING RECORD:** | | | | | | | | **DATE:** | | |
| **Plate MANUFACTURER / PRODUCER:** | | | | | | | | **PRIME CONTRACTOR:** | | |
| Name: | | | | | | | | Name: | | |
| Address: | | | | | | | | Address: | | |
| POC: | | | | | | | | POC: | | |
| Phone No: | | | | | | | | Phone No: | | |
| Fax No: | | | | | | | | Fax No: | | |
| **SPECIFICATION:** MIL-DTL-32262 (MR) | | | | | | | | **REVISION: A** | | **AMENDMENT:** |
| **CONTRACT NO:** | | | | | | | | TECOM PROJECT NO: | | |
| **DCAS REGION:** | | | | | | | | BALLISTIC TEST CONTRACT NO: | | |
| **TEST ITEM IDENTIFICATION:** | | | | | | | | | | |
| Lot No.<br>Class: | | | Plate No. | | | | | Ordered Thickness | | Alloy and Temper |
| **PURPOSE:** ___ Acceptance ___ First Article ___ Development | | | | | | | | | | |
| **SAMPLE:** ___ Primary ___ Retest (Firing Record No. of Failed Sample_____) | | | | | | | | | | |
| **CHEMICAL ANALYSIS:** | | | | | | | | | Other, max. | |
| Si | Fe | Cu | Mn | Mg | Cr | Zn | Ti | | Each | Total | Al: Rem |
| | | | | | | | | | | |
| **MECHANICAL PROPERTIES: Longitudinal (L-orientation) direction** | | | | | | | | | | |
| UTS (ksi): | | | 0.2% YS (ksi): | | | | | Elongation (%): | | |
| **WELDABILITY:** Thickness: Striking Velocity: Required: Actual: Pass / Fail<br>Thickness: Striking Velocity: Required: Actual: Pass / Fail | | | | | | | | | | |
| **BALLISTIC TEST RESULTS:** | | | | | | | | | | |
| Test | Projectile | Obl. (deg) | Actual Thickness (in) | Required $V_{50}$ (fps) | Actual $V_{50}$ (fps) | Pass/ Fail | Notes | | | |
| | | | | | | | | | | |
| **LOTS REPRESENTED BY:** | | | Reduced Testing | | | Audit Testing | | | | |
| Lot [ met ] [failed to meet ] the ballistic requirements of specification MIL-DTL-32262A (MR). | | | | | | | | | | |
| Government Representative | | | Date | | Supplier Representative | | | Date | | |

Figure 1. Aluminum Armor Test Data Form



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3.10.1 Surface and Edge Condition. The surface and edge condition of the plate shall be free of surface cracks, edge cracks, and edge laminations as defined in the contract or purchase order (see 6.2).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Lot. A lot shall consist of all plate of the same alloy; same source of molten metal representing the ingots poured, and ordered thickness which has been processed together by the same mill practice. Unless otherwise specified in the contract or purchase order (see 6.2), the weight of the finished plate in the lot shall not exceed 50,000 pounds and shall be submitted for inspection as a unit.

4.3 First article inspection. When specified in the contract or purchase order, (see 6.2) and before production has commenced, samples of specified material shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3.1. The approval of the first article samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article samples shall be produced using the mill production practice proposed for use on production.

4.3.1 First article inspection, First article inspection except as otherwise indicated in this specification, shall utilize the same requirements and test methods as the conformance inspection or production acceptance inspection shown in 4.4.

4.4 Conformance inspection. Conformance inspection or production acceptance inspection shall include the examination of 4.6 and the tests of 4.7.

4.5 Sampling.

4.5.1 First article inspection.

4.5.1.1 Chemical composition. One (1) sample for chemical analysis shall be removed from each plate selected for ballistic testing and shall meet the requirements of 3.2 when tested as specified in 4.7.1.

4.5.1.2 Mechanical properties. One tension test specimen shall be removed from each plate that has been selected for ballistic testing and shall meet the requirements when tested as specified in 4.7.2.



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4.5.1.3 Ballistic tests. Two plates, 12 inches by 36 inches of each thickness to be supplied on the contract, shall be submitted for ballistic testing in accordance with Appendix B. The orientation of these plates with respect to the rolling direction shall be at the option of the producer (see 6.2).

4.5.2 Conformance inspection.

4.5.2.1 Chemical composition. The sample shall meet the chemical composition requirements of 3.2 when tested as specified in 4.7.1.

4.5.2.1.1 Ingot analysis. At least one sample shall be taken from the molten metal representing the ingots poured as a unit from the same source molten metal. Complete ingot analysis records shall be available to the Government at the producer's facility.

4.5.2.1.2 Product analysis. When sampling has not been made in accordance with 4.5.2.1.1, one sample shall be randomly taken for each 4,000 pounds or less in a lot. Complete product analysis records shall be available to the Government at the contractor's facility.

4.5.2.2 Mechanical properties. Samples for tension tests shall be selected from each lot in accordance with Table IV. Each sample shall be randomly selected from a different plate in the lot, and only one tension test specimen shall be made from each sample. The sample shall meet the requirements when tested as specified in 4.7.2.

TABLE IV. Number of tension tests.

| Lot size, pounds | Minimum number of samples 1/ |
|---|---|
| To 8,000, incl. | 2 |
| 8,001 to 12,000, incl. | 3 |
| 12,001 to 20,000, incl. | 4 |
| 20,001 up | 5 |

1/ If a lot consists of only one plate, one sample shall be required.

4.5.2.3 Ballistic testing. One plate, 12 inches by 36 inches, shall be randomly selected from each lot for ballistic testing. The orientation of the plate with respect to the rolling direction shall be at the option of the producer (see 6.2). The sample shall meet the requirements when tested as specified in 4.7.3.

4.6 Examination.

4.6.1 Visual. Each plate shall be examined for compliance with the identification marking (see 3.8) and workmanship (see 3.10) requirements.

4.6.2 Dimensions. Plates within a lot shall be measured to determine compliance with requirements of paragraph 3.7 in accordance with the sampling procedures approved by the procuring activity and as specified in the contract or purchase order (see 6.2). At a minimum, one plate per lot shall be randomly selected for dimensional inspection.



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4.7 Test specimens.

4.7.1 Chemical composition. Samples for chemical analysis shall be prepared and tested in accordance with one or more ASTM methods of E34, E716, E227, and E1251. In case of dispute, analysis by method E34 shall be the basis for acceptance or rejection.

4.7.2 Mechanical properties. Tension test specimens shall be prepared and tested in accordance with ASTM B557. Specimens shall be taken in the longitudinal (L) direction. For material less than 0.500 inch in thickness, a standard rectangular tension test specimen shall be used. For plate in nominal thickness 0.500 to 1.500 inches, inclusive, tension test specimens shall be taken with the axis midway between the two plate surfaces. For plate in nominal thickness greater than 1.500 inches, the axis of the tension test specimen shall be three-fourths of the distance from one surface to the other.

4.7.3 Ballistic testing. The ordered thickness specified in the contract shall be used to determine the test projectile in accordance with Table V. Ballistic testing shall be in accordance with Appendix B. Test plate thickness, as measured by the ballistic testing agency, shall be used in conjunction with Table V and Appendix B to determine the required $V_{50}$ protection ballistic limit for that plate. Thickness shall be determined as the average of at least four thickness measurements read on a deep throat micrometer or by means of an ultrasonic device to the nearest 0.001 of an inch and rounded off to the nearest 0.005 of an inch. Measurements shall be made on the intended impact area. In those cases where the BL(P) is within ±10 fps of the minimum required value for the measured average thickness (to the nearest 0.005-inch), an interpolation of the appropriate ballistic limit table shall be performed. The average plate thickness, computed to the nearest 0.001-inch, shall be used to determine the minimum required BL(P) for that plate.

TABLE V. Acceptance ballistic test plates.

| Ordered Thickness, Inches | Projectile | Angle of Obliquity in Degrees | TABLE | CLASS I (6061) | CLASS II (6055) |
|---|---|---|---|---|---|
| 0.500 – 0.749 | Cal. .30 AP M2 | 30 | B-I | LISTED | LISTED |
| 0.750 – 0.950 | Cal. .50 FSP | 0 | B-II | N/A | LISTED |
| 0.750 – 1.000 | Cal. .50 FSP | 0 | B-III | LISTED | N/A |
| 1.001 – 1.700 1/ | 20-mm FSP | 0 | B-IV | LISTED | N/A |
| 0.951 – 1.500 2/ | 20-mm FSP | 0 | B-V | N/A | LISTED |
| 1.001 – 1.700 1/ | Cal. .30 AP M2 | 0 | B-VI | LISTED | N/A |
| 0.951 - 1.500 2/ | Cal. .30 AP M2 | 0 | B-VII | N/A | LISTED |
| 1.701 – 2.000 | Cal. .30 AP M2 | 0 | B-VI | LISTED | N/A |
| 1.501 – 3.000 | Cal. .50 AP M2 | 0 | B-VIII | N/A | LISTED |
| 2.001 – 3.000 | Cal. .50 AP M2 | 0 | B-IX | LISTED | N/A |

1/ Two (2) types of projectiles for the same thickness range.
2/ Two (2) types of projectiles for the same thickness range.



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4.7.3.1 Ballistic testing facility. Unless otherwise specified in the contract or purchase orders (see 6.2), the ballistic test plates shall be forwarded to the Commander, U.S. Army Aberdeen Test Center, 400 Colleran Road. Bldg. 358, ATTN: CSTE-DTC-AT-SL-V (K. Beavers), Armor Acceptance – B690, Aberdeen Proving Ground, MD 21005-5059 for ballistic testing for first article or lot acceptance.

4.7.3.2 Incomplete penetrations. When a complete penetration cannot be obtained, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a $V_{50}$ (which obviously cannot be explicitly determined) above the minimum requirement.

4.8 Rejection and retest. Unless otherwise specified in the contract or order (see 6.2) and except as specified in 4.7.2 and 4.7.3, rejection and retest shall be conducted in accordance with 4.8.1, 4.8.1.1, and 4.8.2.

4.8.1 Rejection of first article plates. When one or more first article test specimens fail to meet the requirements of 4.3, the product lot and process, represented by the test plates or specimens shall be subject to rejection except as otherwise provided in a sampling plan approved by the procuring activity and in requirements of 4.8.1.1.

4.8.1.1 Retest of first article samples. Resubmission and retest of first article samples shall not be made until the manufacturer has made necessary corrections in the processing of the material to the satisfaction of the procuring activity. If one of the retest specimens fails the lot shall be permanently rejected with no further testing permitted.

4.8.2 Ballistic. Rejection and retest of ballistic test plates shall be in accordance with B.5.2.

4.9 Reduced testing. At the discretion of the procuring activity and as specified in the contract or purchase order (see 6.2), the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product meeting the testing requirements is being produced and providing the manufacturer agrees to maintain the same manufacturing procedures. Testing for a given plate thickness shall return to standard (non-reduced testing) conditions of one plate per lot, whenever a ballistic test plate fails to meet ballistic requirements.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel components are to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.



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## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The armor specified herein is intended for use on combat vehicles to protect the occupants against small arms fire, fragments, and shrapnel. Application for the AA6061 alloy is the M1114 Up-Armored HMMWV vehicle. Program Executive Offices (PEO) for the Ground Combat Systems will also use AA6061 alloys for their applications. Applications for the AA6055 alloy are as a replacement for AA6061 where increased AP ballistic performance or weldability is required.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Specify classification of alloy, and ordered thickness (see 1.1 and 1.3)
(c) When first article is required (see 3.1 and 4.3).
(d) Special mechanical properties and ballistic requirements if different (see 3.3).
(e) Dimension and tolerance requirements if different (see 3.7 and 3.7.1).
(f) If markings are different and when impression stamping is permissible (see 3.8).
(g) Define surface cracks, edge cracks and edge laminations for the specific application (see 3.10.1)
(h) If the weight of finished plate can exceed 50,000 pounds (see 4.2).
(i) The orientation of the ballistic plate with respect to rolling is different (see 4.5.1.3 and 4.5.2.3).
(j) Dimensional sampling procedure approved by the procuring activity (see 4.6.2).
(k) If approval was requested and received for a different ballistic testing facility (see 4.7.3.1)
(l) Rejection and retest requirement, if different (see 4.8).
(m) If reduced testing is allowed (see 4.9).
(n) Packaging requirements (see 5.1).
(o) Striking velocities for those thicknesses not covered (see A.3.6.1.1).

6.3 Weldability. The Army Research Lab has conducted testing and found that Class II, AA6055 alloys can be welded and demonstrates good shock resistance when tested with the method described in the Ground Combat Vehicle Welding Code, TACOM Drawing 19207-12472301. However, 19207-12472301 does not currently address this alloy. Appendix A provides the additional details to permit welding qualification for armor applications. The fabricator performing welding on these alloys for armor applications is responsible for developing and qualifying welding procedures in accordance with 19207-12472301. Appendix A contains the supplemental information for the welding code. The thickness of the weldments specified in the contract or drawing will be used to determine the proofing projectile and striking velocity in accordance with Appendix A. Tests have demonstrated that material covered by this specification is weldable to itself and other weldable 6000 series alloys using a 4000 series filler and is weldable to 2000 series alloys using a 4000 series filler and to 5000 and 7000 series alloys using a 5000 series filler. It is not weldable to 2000 series alloys by conventional welding techniques. Any



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deviations from these conditions should be demonstrated prior to acceptance by the procuring activity.

6.4 Weld wire. Butts joints were successfully welded with AA4943 weld wire for Class II (AA6055) material when tested with the required proofing projectile per the Ground Combat Vehicle Welding Code 19207-12472301 (see 1.2).

6.5 Density. The density of AA6061 is 0.0975 lb/in³ (2.70 g/cm$^3$). The density of AA6055 is 0.098 lb/in$^3$ (2.72 g/cm$^3$).

6.6 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

| English | Multiply by | Equals | Metric SI unit |
|---|---|---|---|
| inch | 0.0254 | = | meter (m) |
| foot | 0.3048 | = | meter (m) |
| pound | 0.4536 | = | kilogram (kg) |
| foot-lb | 1.3558 | = | joule (j) |
| feet/second | 0.3048 | = | meter per second (m/s) |
| pounds/sq. inch | 0.00689 | = | Mega Pascal (MPa) |
| pound/cubic inch | 27.6799 | = | gram (g) per cubic centimeter (cm) |

6.7 Subject term (key word) listing.

Ballistic testing
Caliber .30 AP M2
Caliber .50 AP M2
Caliber .50 FSP
M1114 HMMWV
Military vehicles
Stress corrosion
20-mm FSP

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



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# ARMOR PLATE, ALUMINUM ALLOY 6055 WELDABLE & ALLOY 6061, UNWELDABLE APPLIQUE

## A.1 SCOPE

A.1.1 Scope. This appendix covers the minimum welding acceptable requirements (striking velocities) of aluminum alloy armor plate, AA6055 when welded and tested in accordance with the provisions specified in the applicable contract. When the material specified by this specification is to be used in a welded armor application the requirements for weldability should specify the Ground Combat Vehicle Welding Code (GCVWC) - Aluminum 19207-12472301. In lieu of this document (Welding Code) or a welding standard for armor and non-armor applications this appendix and the information contained herein is intended for information only. Until the Welding Code or a welding standard for armor and non-armor applications is updated or revised the information contained herein should be referenced for compliance.

## A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

| | | |
|---|---|---|
| TACOM Drawing 19207-12472301 | - | Ground Combat Vehicle Welding Code -- Aluminum, Section 10 |

(Copies of this document are available online at Ground Combat Vehicle Welding Code - Aluminum, dated 6/03/2003 Drawing Number 12472301).

## A.3 REQUIREMENTS

A.3.1 Base Material M Number. Alloy 6055 shall be classified as M23 materials.

A3.2 Tensile Strength of Welded Aluminum Alloys (Table 5.1 of 19207-12472301). The minimum tensile strength of 6055, Class II welds, shall be 24 ksi.

A3.3 Base Metal – Non Critical Welds. Material conforming to this specification is permitted for non-critical welds (paragraph 8.2 of 19207-12472301).

A3.4 Base Metal - Critical Welds. Material conforming to this specification is permitted for critical welds (paragraph 9.2 of 19207-12472301).



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A3.5 Base Metal - Ballistic Welds. Material conforming to this specification is permitted for ballistic welds (paragraph 10.2 of 19207-12472301).

A.3.6 Ballistic Shock (Weldments) test. Testing shall be in accordance with TACOM Drawing 19207-12472301, Ground Combat Vehicle Welding Code -- Aluminum, Section 10, except that nothing in that drawing shall be construed to supersede or invalidate the requirements of this specification.

A.3.6.1 Striking Velocities.

A.3.6.1.1 Flat Weldments (Butt & Corner joints). Striking velocities and proofing projectile shall be chosen from table A-I based on the alloy and thickness of the plate to be impacted. Striking velocities on thicknesses not covered by this table shall be as agreed upon in the contract or purchase order (see 6.2).

A.3.6.2 Number of Impacts. Two fair impacts shall be required on a single welded butt joint. Should an additional welded butt joint be required, two fair impacts shall be required on that additional joint.

TABLE A-I. Minimum required striking velocities for Flat Weldments (Butt & Corner joints)

| **ORDERED THICKNESS (Inches)** | **Plate Proofing Projectile** | **Class II (AA6055)** |
|---|---|---|
| | | |
| 1.000 | 57-mm Aluminum, M1001A | TBD [1/] |
| | | |
| 1.500 | 75-mm Aluminum, M1002A | 1000 fps |

[1/] To be determined



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# ARMOR PLATE, ALUMINUM ALLOY 6055 WELDABLE & ALLOY 6061, UNWELDABLE APPLIQUE

## B.1 SCOPE

B.1.1 Scope. This appendix covers the minimum ballistic limits for acceptable requirements of aluminum alloy armor plate, AA6061 and AA6055 when tested in accordance with the provisions of this specification. When there is mutual agreement between contractor and procuring activity, this appendix becomes a mandatory part of this specification and the information contained herein is intended for compliance.

## B.2 APPLICABLE DOCUMENTS

B.2.1 Government documents.

B.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents are those cited in the solicitation or contract.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-662 - V50 Ballistic Test for Armor

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

## B.3 DEFINITIONS

B.3.1 Complete penetration, (CP). A complete penetration occurs when the impacting projectile, or any fragment thereof, or any fragment of the test specimen perforates the witness plate, resulting in a crack or hole which permits light passage when a 60-watt, 110-volt bulb is placed proximate to the witness plate.

B.3.2 Fair impact. An impact may be considered fair when an un-yawed fragment simulator or test projectile strikes an unsupported area of the target material at a specified obliquity at a distance of at least two projectile diameters from any previous impact or disturbed area resulting from an impact, or from any crack, or from any edge of the test specimen.

B.3.3 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.



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B.3.4 Partial penetration, (PP). Any impact which is not a complete penetration may be considered a partial penetration.

B.3.5 Witness plate. A thin sheet located behind and parallel to the ballistic test sample which is used to detect penetrating projectiles or spall.

B.4 REQUIREMENTS

B.4.1 Resistance to penetration. The minimum required V50 ballistic limit shall be in accordance with the values shown in tables B-I through B-IX.

B.5 TESTS

B.5.1 Ballistic tests. Testing shall be in accordance with MIL-STD-662, V50 Ballistic Test for Armor, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

B.5.1.1 Temperature Conditioning. Prior to the test, the test item(s) shall be temperature conditioned at least eight hours. Thermostatic control shall be such that the average temperature of the item during the test shall be 72 ± 15°F (22 ± 8°C).

B.5.1.2 Protection ballistic limit, BL(P).

B.5.1.2.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) shall be reported.

B.5.1.2.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP[P] velocity being higher than the low CP[P] velocity) exceeds 90 fps, the firing data shall be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit shall be computed on the basis of 4- or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing shall continue until a 10-round ballistic limit has been attained using the smallest possible velocity spread. Ten-round ballistic limits shall be reported as agreed upon between the contractor and procuring activity.

B.5.1.2.3 Reduction of large velocity gap in borderline cases. If the ballistic limit, which has been determined, is within ± 10 fps from the minimum allowable ballistic limit and a gap exists which is greater than 25 fps, then another round, or rounds, shall be fired to reduce the gap to 25 fps or less. The ballistic limit shall then be recomputed using the above



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criteria. The recomputed BL(P) shall be reported as the BL(P) of the plate (in borderline cases, a reduction of the gap between the high partial penetration velocity and the low complete velocity should result in a better evaluation of the BL(P)).

B.5.2 Rejection and retest of ballistic plates.

B.5.2.1 First article tests (rejection). Unless otherwise specified in the contract or order, failure of any of the first article test plates to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

B.5.2.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest plates shall be submitted for first article testing, and both tests shall pass; otherwise, the armor material shall be rejected.

B.5.2.3 Acceptance tests (rejection). Unless otherwise specified in the contract or order, failure of a test plate to meet the ballistic requirements indicates failure of the lot; however, the final decision shall depend on the outcome of retests, if submitted.

B.5.2.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer, upon notification of the failure may submit at his own expense two additional test plates from the same lot for ballistic retest. If either of these plates fail the ballistic test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test plates. If either of these plates fail, the lot shall be permanently rejected.

B.5.3 Disposal of ballistic test plates.

B.5.3.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates shall be returned "as is" to the applicant, at his expense, unless the plates were destroyed in testing.

B.5.3.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered part of the lot they represent, and ownership of the test plates passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification are considered part of the lot they represent and remain the property of the producer. The now rejected lot also remains the property of the producer. The failed plates shall be returned, upon request, as in B.5.3.1.



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TABLE B-I. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 30° obliquity.

| Thickness, inches | Required BL(P), fps | | Thickness, inches | Required BL(P), fps | | Thickness, inches | Required BL(P), fps | |
|---|---|---|---|---|---|---|---|---|
| | Class I | Class II | | Class I | Class II | | Class I | Class II |
| 0.460 | 1290 | 1400 | 0.570 | 1452 | 1617 | 0.680 | 1599 | 1809 |
| 0.465 | 1298 | 1410 | 0.575 | 1459 | 1626 | 0.685 | 1605 | 1817 |
| 0.470 | 1305 | 1421 | 0.580 | 1466 | 1635 | 0.690 | 1612 | 1825 |
| 0.475 | 1313 | 1431 | 0.585 | 1473 | 1645 | 0.695 | 1618 | 1833 |
| 0.480 | 1321 | 1442 | 0.590 | 1480 | 1654 | 0.700 | 1624 | 1841 |
| 0.485 | 1328 | 1452 | 0.595 | 1487 | 1663 | 0.705 | 1630 | 1849 |
| 0.490 | 1336 | 1462 | 0.600 | 1494 | 1672 | 0.710 | 1637 | 1857 |
| 0.495 | 1343 | 1472 | 0.605 | 1500 | 1680 | 0.715 | 1643 | 1865 |
| 0.500 1/ | 1351 | 1483 | 0.610 | 1507 | 1689 | 0.720 | 1649 | 1873 |
| 0.505 | 1358 | 1493 | 0.615 | 1514 | 1698 | 0.725 | 1655 | 1881 |
| 0.510 | 1366 | 1503 | 0.620 | 1521 | 1707 | 0.730 | 1661 | 1889 |
| 0.515 | 1373 | 1512 | 0.625 | 1527 | 1716 | 0.735 | 1668 | 1897 |
| 0.520 | 1381 | 1522 | 0.630 | 1534 | 1724 | 0.740 | 1674 | 1905 |
| 0.525 | 1388 | 1532 | 0.635 | 1541 | 1733 | 0.745 | 1680 | 1913 |
| 0.530 | 1395 | 1542 | 0.640 | 1547 | 1741 | 0.749 2/ | 1685 | 1919 |
| 0.535 | 1402 | 1551 | 0.645 | 1554 | 1750 | 0.750 | 1686 | 1920 |
| 0.540 | 1410 | 1561 | 0.650 | 1560 | 1758 | 0.755 | 1692 | 1928 |
| 0.545 | 1417 | 1570 | 0.655 | 1567 | 1767 | 0.760 | 1699 | 1936 |
| 0.550 | 1424 | 1580 | 0.660 | 1573 | 1775 | 0.765 | 1705 | 1944 |
| 0.555 | 1431 | 1589 | 0.665 | 1580 | 1784 | 0.770 | 1711 | 1951 |
| 0.560 | 1438 | 1599 | 0.670 | 1586 | 1792 | 0.775 | 1717 | 1959 |
| 0.565 | 1445 | 1608 | 0.675 | 1593 | 1800 | 0.780 | 1723 | 1966 |

1/ Specification requirements begin for this ordered thickness.
2/ Specification requirements end for this ordered thickness (See Table B-II).

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.



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TABLE B-II. Minimum required ballistic limits - caliber .50 fragment simulating projectiles at 0° obliquity.

| Thickness, inches | BL(P), fps Required BL(P), fps Class II | Thickness, inches | BL(P), fps Required BL(P), fps Class II | Thickness, inches | BL(P), fps Required BL(P), fps Class II |
|---|---|---|---|---|---|
| 0.720 | 1749 | 0.815 | 2095 | 0.910 | 2509 |
| 0.725 | 1766 | 0.820 | 2115 | 0.915 | 2533 |
| 0.730 | 1782 | 0.825 | 2135 | 0.920 | 2557 |
| 0.735 | 1799 | 0.830 | 2155 | 0.925 | 2582 |
| 0.740 | 1817 | 0.835 | 2176 | 0.930 | 2606 |
| 0.745 | 1834 | 0.840 | 2197 | 0.935 | 2631 |
| 0.750 1/ | 1852 | 0.845 | 2218 | 0.940 | 2656 |
| 0.755 | 1869 | 0.850 | 2239 | 0.945 | 2682 |
| 0.760 | 1887 | 0.855 | 2260 | 0.950 2/ | 2707 |
| 0.765 | 1905 | 0.860 | 2282 | 0.955 | 2733 |
| 0.770 | 1923 | 0.865 | 2304 | 0.960 | 2759 |
| 0.775 | 1942 | 0.870 | 2326 | 0.965 | 2786 |
| 0.780 | 1960 | 0.875 | 2348 | 0.970 | 2812 |
| 0.785 | 1979 | 0.880 | 2370 | 0.975 | 2839 |
| 0.790 | 1998 | 0.885 | 2393 | 0.980 | 2866 |
| 0.795 | 2017 | 0.890 | 2416 | 0.985 | 2894 |
| 0.800 | 2036 | 0.895 | 2439 | 0.990 | 2921 |
| 0.805 | 2055 | 0.900 | 2462 | 0.995 | 2949 |
| 0.810 | 2075 | 0.905 | 2486 | 1.000 | 2977 |

1/ Specification requirements begin for this ordered thickness.
2/ Specification requirements end for this ordered thickness (See Table B-III and Table B-IV).

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.



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TABLE B-III. Minimum required ballistic limits - caliber .50 fragment simulating projectiles at 0° obliquity.

| Thickness, inches | Required BL(P), fps Class I | Thickness, inches | Required BL(P), fps Class I | Thickness, inches | Required BL(P), fps Class I |
|---|---|---|---|---|---|
| 0.720 | 1681 | 0.835 | 2046 | 0.950 | 2485 |
| 0.725 | 1696 | 0.840 | 2064 | 0.955 | 2506 |
| 0.730 | 1711 | 0.845 | 2081 | 0.960 | 2527 |
| 0.735 | 1725 | 0.850 | 2099 | 0.965 | 2549 |
| 0.740 | 1740 | 0.855 | 2117 | 0.970 | 2570 |
| 0.745 | 1755 | 0.860 | 2135 | 0.975 | 2592 |
| 0.750 1/ | 1770 | 0.865 | 2153 | 0.980 | 2614 |
| 0.755 | 1785 | 0.870 | 2171 | 0.985 | 2636 |
| 0.760 | 1801 | 0.875 | 2190 | 0.990 | 2658 |
| 0.765 | 1816 | 0.880 | 2209 | 0.995 | 2680 |
| 0.770 | 1832 | 0.885 | 2227 | 1.000 2/ | 2703 |
| 0.775 | 1848 | 0.890 | 2246 | 1.005 | 2726 |
| 0.780 | 1863 | 0.895 | 2265 | 1.010 | 2749 |
| 0.785 | 1879 | 0.900 | 2284 | 1.015 | 2772 |
| 0.790 | 1895 | 0.905 | 2304 | 1.020 | 2795 |
| 0.795 | 1912 | 0.910 | 2323 | 1.025 | 2819 |
| 0.800 | 1928 | 0.915 | 2343 | 1.030 | 2842 |
| 0.805 | 1945 | 0.920 | 2363 | 1.035 | 2866 |
| 0.810 | 1961 | 0.925 | 2383 | 1.040 | 2890 |
| 0.815 | 1978 | 0.930 | 2403 | 1.045 | 2914 |
| 0.820 | 1995 | 0.935 | 2423 | 1.050 | 2939 |
| 0.825 | 2012 | 0.940 | 2444 | 1.055 | 2963 |
| 0.830 | 2029 | 0.945 | 2464 | 1.060 | 2988 |

1/ Specification requirements begin for this ordered thickness.
2/ Specification requirements end for this ordered thickness (See Table B-IV and Table B-VI).

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.



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TABLE B-IV. Minimum required ballistic limits – 20 mm fragment simulating projectiles at 0° obliquity.

| Thickness, inches | Required BL(P), fps Class I | Thickness, inches | Required BL(P), fps Class I | Thickness, inches | Required BL(P), fps Class I |
|---|---|---|---|---|---|
| 0.980 | 1389 | 1.150 | 1664 | 1.320 | 1991 |
| 0.985 | 1397 | 1.155 | 1673 | 1.325 | 2001 |
| 0.990 | 1404 | 1.160 | 1682 | 1.330 | 2012 |
| 0.995 | 1412 | 1.165 | 1691 | 1.335 | 2022 |
| 1.001[1/] | 1420 | 1.170 | 1700 | 1.340 | 2033 |
| 1.005 | 1427 | 1.175 | 1709 | 1.345 | 2043 |
| 1.010 | 1435 | 1.180 | 1718 | 1.350 | 2054 |
| 1.015 | 1443 | 1.185 | 1727 | 1.355 | 2065 |
| 1.020 | 1450 | 1.190 | 1736 | 1.360 | 2076 |
| 1.025 | 1458 | 1.195 | 1745 | 1.365 | 2087 |
| 1.030 | 1466 | 1.200 | 1755 | 1.370 | 2098 |
| 1.035 | 1474 | 1.205 | 1764 | 1.375 | 2109 |
| 1.040 | 1481 | 1.210 | 1773 | 1.380 | 2120 |
| 1.045 | 1489 | 1.215 | 1783 | 1.385 | 2131 |
| 1.050 | 1497 | 1.220 | 1792 | 1.390 | 2142 |
| 1.055 | 1505 | 1.225 | 1801 | 1.395 | 2153 |
| 1.060 | 1513 | 1.230 | 1811 | 1.400 | 2165 |
| 1.065 | 1521 | 1.235 | 1821 | 1.405 | 2176 |
| 1.070 | 1529 | 1.240 | 1830 | 1.410 | 2187 |
| 1.075 | 1537 | 1.245 | 1840 | 1.415 | 2199 |
| 1.080 | 1546 | 1.250 | 1849 | 1.420 | 2210 |
| 1.085 | 1554 | 1.255 | 1859 | 1.425 | 2222 |
| 1.090 | 1562 | 1.260 | 1869 | 1.430 | 2234 |
| 1.095 | 1570 | 1.265 | 1879 | 1.435 | 2245 |
| 1.100 | 1579 | 1.270 | 1889 | 1.440 | 2257 |
| 1.105 | 1587 | 1.275 | 1899 | 1.445 | 2269 |
| 1.110 | 1596 | 1.280 | 1909 | 1.450 | 2281 |
| 1.115 | 1604 | 1.285 | 1919 | 1.455 | 2293 |
| 1.120 | 1613 | 1.290 | 1929 | 1.460 | 2305 |
| 1.125 | 1621 | 1.295 | 1939 | 1.465 | 2317 |
| 1.130 | 1630 | 1.300 | 1949 | 1.470 | 2329 |
| 1.135 | 1638 | 1.305 | 1959 | 1.475 | 2341 |
| 1.140 | 1647 | 1.310 | 1970 | 1.480 | 2353 |
| 1.145 | 1656 | 1.315 | 1980 | 1.485 | 2366 |

[1/] Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.



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TABLE B-IV. Minimum required ballistic limits – 20 mm fragment simulating projectiles at 0° obliquity - (continued).

| Thickness, inches | Required BL(P), fps Class I | Thickness, inches | Required BL(P), fps Class I | Thickness, inches | Required BL(P), fps Class I |
|---|---|---|---|---|---|
| 1.490 | 2378 | 1.570 | 2585 | 1.650 | 2809 |
| 1.495 | 2390 | 1.575 | 2598 | 1.655 | 2823 |
| 1.500 | 2403 | 1.580 | 2612 | 1.660 | 2838 |
| 1.505 | 2415 | 1.585 | 2625 | 1.665 | 2853 |
| 1.510 | 2428 | 1.590 | 2639 | 1.670 | 2867 |
| 1.515 | 2441 | 1.595 | 2653 | 1.675 | 2882 |
| 1.520 | 2453 | 1.600 | 2666 | 1.680 | 2897 |
| 1.525 | 2466 | 1.605 | 2680 | 1.685 | 2912 |
| 1.530 | 2479 | 1.610 | 2694 | 1.690 | 2928 |
| 1.535 | 2492 | 1.615 | 2708 | 1.695 | 2943 |
| 1.540 | 2505 | 1.620 | 2722 | 1.700 [2/] | 2958 |
| 1.545 | 2518 | 1.625 | 2737 | 1.705 | 2973 |
| 1.550 | 2531 | 1.630 | 2751 | 1.710 | 2989 |
| 1.555 | 2544 | 1.635 | 2765 | 1.715 | 3004 |
| 1.560 | 2558 | 1.640 | 2780 | 1.720 | 3019 |
| 1.565 | 2571 | 1.645 | 2794 | 1.725 | 3035 |

[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.



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TABLE B-V. Minimum required ballistic limits – 20 mm fragment simulating projectiles at 0° obliquity.

| Thickness, inches | Required BL(P), fps Class II | Thickness, inches | Required BL(P), fps Class II | Thickness, inches | Required BL(P), fps Class II |
|---|---|---|---|---|---|
| 0.930 | 1229 | 1.090 | 1509 | 1.255 | 1863 |
| 0.935 | 1237 | 1.095 | 1518 | 1.260 | 1875 |
| 0.940 | 1245 | 1.100 | 1528 | 1.265 | 1887 |
| 0.945 | 1253 | 1.105 | 1538 | 1.270 | 1900 |
| 0.950 | 1261 | 1.110 | 1548 | 1.275 | 1912 |
| 0.951 1/ | 1263 | 1.115 | 1558 | 1.280 | 1924 |
| 0.955 | 1269 | 1.120 | 1568 | 1.285 | 1936 |
| 0.960 | 1277 | 1.125 | 1578 | 1.290 | 1949 |
| 0.965 | 1286 | 1.130 | 1588 | 1.295 | 1961 |
| 0.970 | 1294 | 1.135 | 1598 | 1.300 | 1974 |
| 0.975 | 1302 | 1.140 | 1608 | 1.305 | 1987 |
| 0.980 | 1310 | 1.145 | 1619 | 1.310 | 1999 |
| 0.985 | 1319 | 1.150 | 1629 | 1.315 | 2012 |
| 0.990 | 1327 | 1.155 | 1640 | 1.320 | 2025 |
| 0.995 | 1336 | 1.160 | 1650 | 1.325 | 2038 |
| 1.000 | 1344 | 1.165 | 1661 | 1.330 | 2051 |
| 1.005 | 1353 | 1.170 | 1671 | 1.335 | 2064 |
| 1.010 | 1362 | 1.175 | 1682 | 1.340 | 2078 |
| 1.015 | 1371 | 1.180 | 1693 | 1.345 | 2091 |
| 1.020 | 1379 | 1.185 | 1704 | 1.350 | 2104 |
| 1.025 | 1388 | 1.190 | 1715 | 1.355 | 2118 |
| 1.030 | 1397 | 1.195 | 1726 | 1.360 | 2131 |
| 1.035 | 1406 | 1.200 | 1737 | 1.365 | 2145 |
| 1.040 | 1415 | 1.205 | 1748 | 1.370 | 2159 |
| 1.045 | 1424 | 1.210 | 1759 | 1.375 | 2173 |
| 1.050 | 1433 | 1.215 | 1770 | 1.380 | 2187 |
| 1.055 | 1443 | 1.220 | 1782 | 1.385 | 2201 |
| 1.060 | 1452 | 1.225 | 1793 | 1.390 | 2215 |
| 1.065 | 1461 | 1.230 | 1805 | 1.395 | 2229 |
| 1.070 | 1470 | 1.235 | 1816 | 1.400 | 2243 |
| 1.075 | 1480 | 1.240 | 1828 | 1.405 | 2258 |
| 1.080 | 1489 | 1.245 | 1840 | 1.410 | 2272 |
| 1.085 | 1499 | 1.250 | 1852 | 1.415 | 2287 |

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.



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TABLE B-V. Minimum required ballistic limits – 20 mm fragment simulating projectiles at 0° obliquity - (continued).

| Thickness, inches | Required BL(P), fps Class II | Thickness, inches | Required BL(P), fps Class II | Thickness, inches | Required BL(P), fps Class II |
|---|---|---|---|---|---|
| 1.420 | 2302 | 1.455 | 2407 | 1.490 | 2517 |
| 1.425 | 2316 | 1.460 | 2422 | 1.495 | 2533 |
| 1.430 | 2331 | 1.465 | 2438 | 1.500 2/ | 2550 |
| 1.435 | 2346 | 1.470 | 2454 | 1.505 | 2566 |
| 1.440 | 2361 | 1.475 | 2469 | 1.510 | 2583 |
| 1.445 | 2376 | 1.480 | 2485 | 1.515 | 2599 |
| 1.450 | 2392 | 1.485 | 2501 | 1.520 | 2616 |

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.



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TABLE B-VI. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity.

| Thickness, inches | Required BL(P), fps Class I | Thickness, inches | Required BL(P), fps Class I | Thickness, inches | Required BL(P), fps Class I |
|---|---|---|---|---|---|
| 0.980 | 1792 | 1.150 | 1994 | 1.320 | 2177 |
| 0.985 | 1798 | 1.155 | 2000 | 1.325 | 2183 |
| 0.990 | 1804 | 1.160 | 2005 | 1.330 | 2188 |
| 0.995 | 1810 | 1.165 | 2011 | 1.335 | 2193 |
| 1.001 1/ | 1817 | 1.170 | 2016 | 1.340 | 2198 |
| 1.005 | 1823 | 1.175 | 2022 | 1.345 | 2203 |
| 1.010 | 1829 | 1.180 | 2027 | 1.350 | 2208 |
| 1.015 | 1836 | 1.185 | 2033 | 1.355 | 2213 |
| 1.020 | 1842 | 1.190 | 2038 | 1.320 | 2177 |
| 1.025 | 1848 | 1.195 | 2044 | 1.325 | 2183 |
| 1.030 | 1854 | 1.200 | 2050 | 1.330 | 2188 |
| 1.035 | 1860 | 1.205 | 2055 | 1.335 | 2193 |
| 1.040 | 1866 | 1.210 | 2060 | 1.340 | 2198 |
| 1.045 | 1872 | 1.215 | 2066 | 1.345 | 2203 |
| 1.050 | 1878 | 1.220 | 2071 | 1.350 | 2208 |
| 1.055 | 1884 | 1.225 | 2077 | 1.355 | 2213 |
| 1.060 | 1890 | 1.230 | 2082 | 1.360 | 2218 |
| 1.065 | 1896 | 1.235 | 2088 | 1.365 | 2223 |
| 1.070 | 1902 | 1.240 | 2093 | 1.370 | 2229 |
| 1.075 | 1907 | 1.245 | 2098 | 1.375 | 2234 |
| 1.080 | 1913 | 1.250 | 2104 | 1.380 | 2239 |
| 1.085 | 1919 | 1.255 | 2109 | 1.385 | 2244 |
| 1.090 | 1925 | 1.260 | 2114 | 1.390 | 2249 |
| 1.095 | 1931 | 1.265 | 2120 | 1.395 | 2254 |
| 1.100 | 1937 | 1.270 | 2125 | 1.400 | 2259 |
| 1.105 | 1942 | 1.275 | 2130 | 1.405 | 2264 |
| 1.110 | 1948 | 1.280 | 2136 | 1.410 | 2269 |
| 1.115 | 1954 | 1.285 | 2141 | 1.415 | 2273 |
| 1.120 | 1960 | 1.290 | 2146 | 1.420 | 2278 |
| 1.125 | 1965 | 1.295 | 2151 | 1.425 | 2283 |
| 1.130 | 1971 | 1.300 | 2157 | 1.430 | 2288 |
| 1.135 | 1977 | 1.305 | 2162 | 1.435 | 2293 |
| 1.140 | 1983 | 1.310 | 2167 | 1.440 | 2298 |
| 1.145 | 1988 | 1.315 | 2172 | 1.445 | 2303 |

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.



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TABLE B-VI. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity (continued).

| Thickness, inches | Required BL(P), fps Class I | Thickness, inches | Required BL(P), fps Class I | Thickness, inches | Required BL(P), fps Class I |
|---|---|---|---|---|---|
| 1.450 | 2308 | 1.625 | 2473 | 1.800 | 2629 |
| 1.455 | 2313 | 1.630 | 2478 | 1.805 | 2633 |
| 1.460 | 2318 | 1.635 | 2482 | 1.810 | 2637 |
| 1.465 | 2322 | 1.640 | 2487 | 1.815 | 2641 |
| 1.470 | 2327 | 1.645 | 2492 | 1.820 | 2646 |
| 1.475 | 2332 | 1.650 | 2496 | 1.825 | 2650 |
| 1.480 | 2337 | 1.655 | 2501 | 1.830 | 2654 |
| 1.485 | 2342 | 1.660 | 2505 | 1.835 | 2658 |
| 1.490 | 2347 | 1.665 | 2510 | 1.840 | 2663 |
| 1.495 | 2352 | 1.670 | 2514 | 1.845 | 2667 |
| 1.500 | 2356 | 1.675 | 2519 | 1.850 | 2671 |
| 1.505 | 2361 | 1.680 | 2523 | 1.855 | 2675 |
| 1.510 | 2366 | 1.685 | 2528 | 1.860 | 2680 |
| 1.515 | 2371 | 1.690 | 2532 | 1.865 | 2684 |
| 1.520 | 2376 | 1.695 | 2537 | 1.870 | 2688 |
| 1.525 | 2380 | 1.700 | 2541 | 1.875 | 2692 |
| 1.530 | 2385 | 1.705 | 2545 | 1.880 | 2697 |
| 1.535 | 2390 | 1.710 | 2550 | 1.885 | 2701 |
| 1.540 | 2394 | 1.715 | 2554 | 1.890 | 2705 |
| 1.545 | 2399 | 1.720 | 2559 | 1.895 | 2709 |
| 1.550 | 2404 | 1.725 | 2563 | 1.900 | 2713 |
| 1.555 | 2409 | 1.730 | 2568 | 1.905 | 2717 |
| 1.560 | 2413 | 1.735 | 2572 | 1.910 | 2722 |
| 1.565 | 2418 | 1.740 | 2576 | 1.915 | 2726 |
| 1.570 | 2423 | 1.745 | 2581 | 1.920 | 2730 |
| 1.575 | 2427 | 1.750 | 2585 | 1.925 | 2734 |
| 1.580 | 2432 | 1.755 | 2590 | 1.930 | 2738 |
| 1.585 | 2437 | 1.760 | 2594 | 1.935 | 2742 |
| 1.590 | 2441 | 1.765 | 2598 | 1.940 | 2746 |
| 1.595 | 2446 | 1.770 | 2603 | 1.945 | 2751 |
| 1.600 | 2450 | 1.775 | 2607 | 1.950 | 2755 |
| 1.605 | 2455 | 1.780 | 2611 | 1.955 | 2759 |
| 1.610 | 2460 | 1.785 | 2616 | 1.960 | 2763 |
| 1.615 | 2464 | 1.790 | 2620 | 1.965 | 2767 |
| 1.620 | 2469 | 1.795 | 2624 | 1.970 | 2771 |



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TABLE B-VI. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity (continued).

| Thickness, inches | Required BL(P), fps Class I | Thickness, inches | Required BL(P), fps Class I | Thickness, inches | Required BL(P), fps Class I |
|---|---|---|---|---|---|
| 1.975 | 2775 | 2.005 | 2800 | 2.035 | 2824 |
| 1.980 | 2779 | 2.010 | 2804 | 2.040 | 2828 |
| 1.985 | 2783 | 2.015 | 2808 | 2.045 | 2832 |
| 1.990 | 2787 | 2.020 | 2812 | 2.050 | 2836 |
| 1.995 | 2791 | 2.025 | 2816 | 2.055 | 2840 |
| 2.000 2/ | 2796 | 2.030 | 2820 | 2.060 | 2844 |

2/ Specification requirements end for this ordered thickness (See Table B-IX).

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.



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TABLE B-VII. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity.

| Thickness, inches | Required BL(P), fps Class II | Thickness, inches | Required BL(P), fps Class II | Thickness, inches | Required BL(P), fps Class II |
|---|---|---|---|---|---|
| 0.930 | 1866 | 1.090 | 2080 | 1.255 | 2280 |
| 0.935 | 1873 | 1.095 | 2086 | 1.260 | 2285 |
| 0.940 | 1880 | 1.100 | 2092 | 1.265 | 2291 |
| 0.945 | 1887 | 1.105 | 2099 | 1.270 | 2297 |
| 0.950 | 1894 | 1.110 | 2105 | 1.275 | 2303 |
| 0.951 1/ | 1895 | 1.115 | 2111 | 1.280 | 2308 |
| 0.955 | 1901 | 1.120 | 2117 | 1.285 | 2314 |
| 0.960 | 1908 | 1.125 | 2124 | 1.290 | 2320 |
| 0.965 | 1914 | 1.130 | 2130 | 1.295 | 2325 |
| 0.970 | 1921 | 1.135 | 2136 | 1.300 | 2331 |
| 0.975 | 1928 | 1.140 | 2142 | 1.305 | 2337 |
| 0.980 | 1935 | 1.145 | 2148 | 1.310 | 2342 |
| 0.985 | 1942 | 1.150 | 2155 | 1.315 | 2348 |
| 0.990 | 1949 | 1.155 | 2161 | 1.320 | 2354 |
| 0.995 | 1955 | 1.160 | 2167 | 1.325 | 2359 |
| 1.000 | 1962 | 1.165 | 2173 | 1.330 | 2365 |
| 1.005 | 1969 | 1.170 | 2179 | 1.335 | 2370 |
| 1.010 | 1976 | 1.175 | 2185 | 1.340 | 2376 |
| 1.015 | 1982 | 1.180 | 2191 | 1.345 | 2382 |
| 1.020 | 1989 | 1.185 | 2197 | 1.350 | 2387 |
| 1.025 | 1995 | 1.190 | 2203 | 1.355 | 2393 |
| 1.030 | 2002 | 1.195 | 2209 | 1.360 | 2398 |
| 1.035 | 2009 | 1.200 | 2215 | 1.365 | 2404 |
| 1.040 | 2015 | 1.205 | 2221 | 1.370 | 2409 |
| 1.045 | 2022 | 1.210 | 2227 | 1.375 | 2415 |
| 1.050 | 2028 | 1.215 | 2233 | 1.380 | 2420 |
| 1.055 | 2035 | 1.220 | 2239 | 1.385 | 2425 |
| 1.060 | 2041 | 1.225 | 2245 | 1.390 | 2431 |
| 1.065 | 2048 | 1.230 | 2250 | 1.395 | 2436 |
| 1.070 | 2054 | 1.235 | 2256 | 1.400 | 2442 |
| 1.075 | 2061 | 1.240 | 2262 | 1.405 | 2447 |
| 1.080 | 2067 | 1.245 | 2268 | 1.410 | 2453 |
| 1.085 | 2073 | 1.250 | 2274 | 1.415 | 2458 |

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.



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TABLE B-VII. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity (continued).

| Thickness, inches | Required BL(P), fps Class II | Thickness, inches | Required BL(P), fps Class II | Thickness, inches | Required BL(P), fps Class II |
|---|---|---|---|---|---|
| 1.420 | 2463 | 1.455 | 2500 | 1.490 | 2537 |
| 1.425 | 2469 | 1.460 | 2506 | 1.495 | 2542 |
| 1.430 | 2474 | 1.465 | 2511 | 1.500 2/ | 2548 |
| 1.435 | 2479 | 1.470 | 2516 | 1.505 | 2553 |
| 1.440 | 2485 | 1.475 | 2522 | 1.510 | 2558 |
| 1.445 | 2490 | 1.480 | 2527 | 1.515 | 2563 |
| 1.450 | 2495 | 1.485 | 2532 | 1.520 | 2568 |

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.



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TABLE B-VIII. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity.

| Thickness, inches | Required BL(P), fps Class II | Thickness, inches | Required BL(P), fps Class II | Thickness, inches | Required BL(P), fps Class II |
|---|---|---|---|---|---|
| 1.460 | 1823 | 1.630 | 1964 | 1.805 | 2099 |
| 1.465 | 1828 | 1.635 | 1968 | 1.810 | 2103 |
| 1.470 | 1832 | 1.640 | 1972 | 1.815 | 2107 |
| 1.475 | 1836 | 1.645 | 1976 | 1.820 | 2111 |
| 1.480 | 1840 | 1.650 | 1980 | 1.825 | 2114 |
| 1.485 | 1845 | 1.655 | 1984 | 1.830 | 2118 |
| 1.490 | 1849 | 1.660 | 1988 | 1.835 | 2122 |
| 1.495 | 1853 | 1.665 | 1992 | 1.840 | 2125 |
| 1.500 | 1857 | 1.670 | 1996 | 1.845 | 2129 |
| 1.501 1/ | 1858 | 1.675 | 2000 | 1.850 | 2133 |
| 1.505 | 1862 | 1.680 | 2004 | 1.855 | 2137 |
| 1.510 | 1866 | 1.685 | 2008 | 1.860 | 2140 |
| 1.515 | 1870 | 1.690 | 2012 | 1.865 | 2144 |
| 1.520 | 1874 | 1.695 | 2016 | 1.870 | 2148 |
| 1.525 | 1878 | 1.700 | 2019 | 1.875 | 2151 |
| 1.530 | 1883 | 1.705 | 2023 | 1.880 | 2155 |
| 1.535 | 1887 | 1.710 | 2027 | 1.885 | 2158 |
| 1.540 | 1891 | 1.715 | 2031 | 1.890 | 2162 |
| 1.545 | 1895 | 1.720 | 2035 | 1.895 | 2166 |
| 1.550 | 1899 | 1.725 | 2039 | 1.900 | 2169 |
| 1.555 | 1903 | 1.730 | 2043 | 1.905 | 2173 |
| 1.560 | 1907 | 1.735 | 2046 | 1.910 | 2177 |
| 1.565 | 1912 | 1.740 | 2050 | 1.915 | 2180 |
| 1.570 | 1916 | 1.745 | 2054 | 1.920 | 2184 |
| 1.575 | 1920 | 1.750 | 2058 | 1.925 | 2187 |
| 1.580 | 1924 | 1.755 | 2062 | 1.930 | 2191 |
| 1.585 | 1928 | 1.760 | 2066 | 1.935 | 2195 |
| 1.590 | 1932 | 1.765 | 2069 | 1.940 | 2198 |
| 1.595 | 1936 | 1.770 | 2073 | 1.945 | 2202 |
| 1.600 | 1940 | 1.775 | 2077 | 1.950 | 2205 |
| 1.605 | 1944 | 1.780 | 2081 | 1.955 | 2209 |
| 1.610 | 1948 | 1.785 | 2084 | 1.960 | 2212 |
| 1.615 | 1952 | 1.790 | 2088 | 1.965 | 2216 |
| 1.620 | 1956 | 1.795 | 2092 | 1.970 | 2219 |
| 1.625 | 1960 | 1.800 | 2096 | 1.975 | 2223 |

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.



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TABLE B-VIII. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity (continued).

| Thickness, inches | Required BL(P), fps Class II | Thickness, inches | Required BL(P), fps Class II | Thickness, inches | Required BL(P), fps Class II |
|---|---|---|---|---|---|
| 1.980 | 2226 | 2.160 | 2350 | 2.340 | 2467 |
| 1.985 | 2230 | 2.165 | 2353 | 2.345 | 2470 |
| 1.990 | 2234 | 2.170 | 2357 | 2.350 | 2474 |
| 1.995 | 2237 | 2.175 | 2360 | 2.355 | 2477 |
| 2.000 | 2241 | 2.180 | 2363 | 2.360 | 2480 |
| 2.005 | 2244 | 2.185 | 2367 | 2.365 | 2483 |
| 2.010 | 2248 | 2.190 | 2370 | 2.370 | 2486 |
| 2.015 | 2251 | 2.195 | 2373 | 2.375 | 2489 |
| 2.020 | 2255 | 2.200 | 2377 | 2.380 | 2493 |
| 2.025 | 2258 | 2.205 | 2380 | 2.385 | 2496 |
| 2.030 | 2261 | 2.210 | 2383 | 2.390 | 2499 |
| 2.035 | 2265 | 2.215 | 2386 | 2.395 | 2502 |
| 2.040 | 2268 | 2.220 | 2390 | 2.400 | 2505 |
| 2.045 | 2272 | 2.225 | 2393 | 2.405 | 2508 |
| 2.050 | 2275 | 2.230 | 2396 | 2.410 | 2511 |
| 2.055 | 2279 | 2.235 | 2400 | 2.415 | 2515 |
| 2.060 | 2282 | 2.240 | 2403 | 2.420 | 2518 |
| 2.065 | 2286 | 2.245 | 2406 | 2.425 | 2521 |
| 2.070 | 2289 | 2.250 | 2409 | 2.430 | 2524 |
| 2.075 | 2292 | 2.255 | 2413 | 2.435 | 2527 |
| 2.080 | 2296 | 2.260 | 2416 | 2.440 | 2530 |
| 2.085 | 2299 | 2.265 | 2419 | 2.445 | 2533 |
| 2.090 | 2303 | 2.270 | 2422 | 2.450 | 2536 |
| 2.095 | 2306 | 2.275 | 2426 | 2.455 | 2539 |
| 2.100 | 2310 | 2.280 | 2429 | 2.460 | 2543 |
| 2.105 | 2313 | 2.285 | 2432 | 2.465 | 2546 |
| 2.110 | 2316 | 2.290 | 2435 | 2.470 | 2549 |
| 2.115 | 2320 | 2.295 | 2439 | 2.475 | 2552 |
| 2.120 | 2323 | 2.300 | 2442 | 2.480 | 2555 |
| 2.125 | 2326 | 2.305 | 2445 | 2.485 | 2558 |
| 2.130 | 2330 | 2.310 | 2448 | 2.490 | 2561 |
| 2.135 | 2333 | 2.315 | 2451 | 2.495 | 2564 |
| 2.140 | 2337 | 2.320 | 2455 | 2.500 | 2567 |
| 2.145 | 2340 | 2.325 | 2458 | 2.505 | 2570 |
| 2.150 | 2343 | 2.330 | 2461 | 2.510 | 2573 |
| 2.155 | 2347 | 2.335 | 2464 | 2.515 | 2576 |



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TABLE B-VIII. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity (continued).

| Thickness, inches | Required BL(P), fps Class II | Thickness, inches | Required BL(P), fps Class II | Thickness, inches | Required BL(P), fps Class II |
|---|---|---|---|---|---|
| 2.520 | 2579 | 2.690 | 2681 | 2.860 | 2779 |
| 2.525 | 2582 | 2.695 | 2684 | 2.865 | 2781 |
| 2.530 | 2585 | 2.700 | 2687 | 2.870 | 2784 |
| 2.535 | 2588 | 2.705 | 2690 | 2.875 | 2787 |
| 2.540 | 2591 | 2.710 | 2692 | 2.880 | 2790 |
| 2.545 | 2594 | 2.715 | 2695 | 2.885 | 2793 |
| 2.550 | 2597 | 2.720 | 2698 | 2.890 | 2795 |
| 2.555 | 2601 | 2.725 | 2701 | 2.895 | 2798 |
| 2.560 | 2604 | 2.730 | 2704 | 2.900 | 2801 |
| 2.565 | 2607 | 2.735 | 2707 | 2.905 | 2804 |
| 2.570 | 2610 | 2.740 | 2710 | 2.910 | 2807 |
| 2.575 | 2613 | 2.745 | 2713 | 2.915 | 2809 |
| 2.580 | 2616 | 2.750 | 2716 | 2.920 | 2812 |
| 2.585 | 2619 | 2.755 | 2719 | 2.925 | 2815 |
| 2.590 | 2622 | 2.760 | 2721 | 2.930 | 2818 |
| 2.595 | 2625 | 2.765 | 2724 | 2.935 | 2821 |
| 2.600 | 2628 | 2.770 | 2727 | 2.940 | 2823 |
| 2.605 | 2631 | 2.775 | 2730 | 2.945 | 2826 |
| 2.610 | 2634 | 2.780 | 2733 | 2.950 | 2829 |
| 2.615 | 2636 | 2.785 | 2736 | 2.955 | 2832 |
| 2.620 | 2639 | 2.790 | 2739 | 2.960 | 2835 |
| 2.625 | 2642 | 2.795 | 2742 | 2.965 | 2837 |
| 2.630 | 2645 | 2.800 | 2744 | 2.970 | 2840 |
| 2.635 | 2648 | 2.805 | 2747 | 2.975 | 2843 |
| 2.640 | 2651 | 2.810 | 2750 | 2.980 | 2846 |
| 2.645 | 2654 | 2.815 | 2753 | 2.985 | 2848 |
| 2.650 | 2657 | 2.820 | 2756 | 2.990 | 2851 |
| 2.655 | 2660 | 2.825 | 2759 | 2.995 | 2854 |
| 2.660 | 2663 | 2.830 | 2762 | 3.000 [2/] | 2857 |
| 2.665 | 2666 | 2.835 | 2764 | 3.005 | 2859 |
| 2.670 | 2669 | 2.840 | 2767 | 3.010 | 2862 |
| 2.675 | 2672 | 2.845 | 2770 | 3.015 | 2865 |
| 2.680 | 2675 | 2.850 | 2773 | 3.020 | 2868 |
| 2.685 | 2678 | 2.855 | 2776 | 3.025 | 2870 |

[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.



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TABLE B-IX. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity.

| Thickness, inches | Required BL(P), fps Class I | Thickness, inches | Required BL(P), fps Class I | Thickness, inches | Required BL(P), fps Class I |
|---|---|---|---|---|---|
| 1.950 | 2041 | 2.130 | 2161 | 2.310 | 2275 |
| 1.955 | 2045 | 2.135 | 2164 | 2.315 | 2278 |
| 1.960 | 2048 | 2.140 | 2168 | 2.320 | 2281 |
| 1.965 | 2051 | 2.145 | 2171 | 2.325 | 2284 |
| 1.970 | 2055 | 2.150 | 2174 | 2.330 | 2287 |
| 1.975 | 2058 | 2.155 | 2177 | 2.335 | 2290 |
| 1.980 | 2062 | 2.160 | 2180 | 2.340 | 2293 |
| 1.985 | 2065 | 2.165 | 2184 | 2.345 | 2296 |
| 1.990 | 2068 | 2.170 | 2187 | 2.350 | 2299 |
| 1.995 | 2072 | 2.175 | 2190 | 2.355 | 2303 |
| 2.001 1/ | 2075 | 2.180 | 2193 | 2.360 | 2306 |
| 2.005 | 2079 | 2.185 | 2196 | 2.365 | 2309 |
| 2.010 | 2082 | 2.190 | 2200 | 2.370 | 2312 |
| 2.015 | 2085 | 2.195 | 2203 | 2.375 | 2315 |
| 2.020 | 2089 | 2.200 | 2206 | 2.380 | 2318 |
| 2.025 | 2092 | 2.205 | 2209 | 2.385 | 2321 |
| 2.030 | 2095 | 2.210 | 2212 | 2.390 | 2324 |
| 2.035 | 2099 | 2.215 | 2216 | 2.395 | 2327 |
| 2.040 | 2102 | 2.220 | 2219 | 2.400 | 2330 |
| 2.045 | 2105 | 2.225 | 2222 | 2.405 | 2333 |
| 2.050 | 2109 | 2.230 | 2225 | 2.410 | 2336 |
| 2.055 | 2112 | 2.235 | 2228 | 2.415 | 2339 |
| 2.060 | 2115 | 2.240 | 2231 | 2.420 | 2342 |
| 2.065 | 2119 | 2.245 | 2234 | 2.425 | 2345 |
| 2.070 | 2122 | 2.250 | 2238 | 2.430 | 2348 |
| 2.075 | 2125 | 2.255 | 2241 | 2.435 | 2351 |
| 2.080 | 2128 | 2.260 | 2244 | 2.440 | 2354 |
| 2.085 | 2132 | 2.265 | 2247 | 2.445 | 2357 |
| 2.090 | 2135 | 2.270 | 2250 | 2.450 | 2360 |
| 2.095 | 2138 | 2.275 | 2253 | 2.455 | 2363 |
| 2.100 | 2142 | 2.280 | 2256 | 2.460 | 2366 |
| 2.105 | 2145 | 2.285 | 2259 | 2.465 | 2369 |
| 2.110 | 2148 | 2.290 | 2263 | 2.470 | 2372 |
| 2.115 | 2151 | 2.295 | 2266 | 2.475 | 2375 |
| 2.120 | 2155 | 2.300 | 2269 | 2.480 | 2378 |
| 2.125 | 2158 | 2.305 | 2272 | 2.485 | 2380 |

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.



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TABLE B-IX. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity (continued).

| Thickness, inches | Required BL(P), fps | Thickness, inches | Required BL(P), fps | Thickness, inches | Required BL(P), fps |
|---|---|---|---|---|---|
| | Class I | | Class I | | Class I |
| 2.490 | 2383 | 2.675 | 2490 | 2.860 | 2593 |
| 2.495 | 2386 | 2.680 | 2493 | 2.865 | 2595 |
| 2.500 | 2389 | 2.685 | 2496 | 2.870 | 2598 |
| 2.505 | 2392 | 2.690 | 2499 | 2.875 | 2601 |
| 2.510 | 2395 | 2.695 | 2501 | 2.880 | 2604 |
| 2.515 | 2398 | 2.700 | 2504 | 2.885 | 2606 |
| 2.520 | 2401 | 2.705 | 2507 | 2.890 | 2609 |
| 2.525 | 2404 | 2.710 | 2510 | 2.895 | 2612 |
| 2.530 | 2407 | 2.715 | 2513 | 2.900 | 2614 |
| 2.535 | 2410 | 2.720 | 2516 | 2.905 | 2617 |
| 2.540 | 2413 | 2.725 | 2518 | 2.910 | 2620 |
| 2.545 | 2416 | 2.730 | 2521 | 2.915 | 2622 |
| 2.550 | 2419 | 2.735 | 2524 | 2.920 | 2625 |
| 2.555 | 2421 | 2.740 | 2527 | 2.925 | 2628 |
| 2.560 | 2424 | 2.745 | 2529 | 2.930 | 2630 |
| 2.565 | 2427 | 2.750 | 2532 | 2.935 | 2633 |
| 2.570 | 2430 | 2.755 | 2535 | 2.940 | 2636 |
| 2.575 | 2433 | 2.760 | 2538 | 2.945 | 2639 |
| 2.580 | 2436 | 2.765 | 2541 | 2.950 | 2641 |
| 2.585 | 2439 | 2.770 | 2543 | 2.955 | 2644 |
| 2.590 | 2442 | 2.775 | 2546 | 2.960 | 2647 |
| 2.595 | 2445 | 2.780 | 2549 | 2.965 | 2649 |
| 2.600 | 2447 | 2.785 | 2552 | 2.970 | 2652 |
| 2.605 | 2450 | 2.790 | 2554 | 2.975 | 2654 |
| 2.610 | 2453 | 2.795 | 2557 | 2.980 | 2657 |
| 2.615 | 2456 | 2.800 | 2560 | 2.985 | 2660 |
| 2.620 | 2459 | 2.805 | 2563 | 2.990 | 2662 |
| 2.625 | 2462 | 2.810 | 2565 | 2.995 | 2665 |
| 2.630 | 2465 | 2.815 | 2568 | 3.000 [2/] | 2668 |
| 2.635 | 2468 | 2.820 | 2571 | 3.005 | 2670 |
| 2.640 | 2470 | 2.825 | 2574 | 3.010 | 2673 |
| 2.645 | 2473 | 2.830 | 2576 | 3.015 | 2676 |
| 2.650 | 2476 | 2.835 | 2579 | 3.020 | 2678 |
| 2.655 | 2479 | 2.840 | 2582 | 3.025 | 2681 |
| 2.660 | 2482 | 2.845 | 2585 | 3.030 | 2684 |
| 2.665 | 2485 | 2.850 | 2587 | 3.035 | 2686 |
| 2.670 | 2487 | 2.855 | 2590 | 3.040 | 2689 |

[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.



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CONCLUDING MATERIAL

Custodians:
Army – MR

Preparing activity:
ARMY – MR
(Project 9535-2014-004)

Review activities:
Army – AR, AT, AV, TE
DLA – IS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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